Harshvardhan Soni

Rajkot, Gujarat +91-9313284692 soniharshvardhan14@gmail.com harshvardhan-soni14

SUMMARY

Having a strong interest in VLSI and digital circuit design. I've worked with Verilog, basic digital testing techniques like path delay and transition fault testing, and have a good understanding of VLSI design and verification concepts. I've also explored microcontrollers and embedded C, but my main focus is now on learning and growing in the field of VLSI, especially in physical design and testing. I'm excited to start my career in the semiconductor industry and apply what I've learned to real-world projects.

PROJECTS

Cabin Crew Lighting System | *Digital Logic Design, Verilog, Model Sim, EDA Playground*

- Developed a digitally controlled Crew Cabin Lighting System simulating real-time modes like Normal, Dimming, and Emergency.
- Designed using **Verilog HDL**, **FSM**, and combinational logic, verified via waveform simulations on Model Sim and EDA Playground, without microcontroller use.

IoT-Based Water Tank Level Monitoring and Controlling System | *STM32, HC-SR04, LoRa RA-01, SPI*

- Developed a contactless water monitoring and motor control system using STM32, ultrasonic sensors, and **LoRa RA-01** (up to 2 km), removing manual checks and Wi-Fi dependency.
- Integrated solar-powered unit, TFT display, relay automation, and manual override, applying Human-Centered Design with real-user feedback from tenement testing.

UART Communication with LED Control | *Embedded C, ARM Cortex M4, UART, GPIO*

- Developed a UART-based LED control system using a microcontroller, ensuring accurate signal transmission with minimal latency.
- Designed embedded firmware in Embedded C, configuring **UART** on **ARM Cortex** for reliable data exchange with external devices

SKILLS

VLSI: Digital Electronics, Verilog, Digital VLSI Testing, Deeds, Multisim
Programming Languages: C, Java, Python, DART
Embedded Systems: ARM Cortex M4, UART, Embedded C, ATmega32, STM32, Proteus, Keil µVision
Software Development: Flutter, JavaScript, HTML, CSS
Extras: CAN Protocol, Automotive Embedded Systems, ADAS, Fusion 360, CircuitVerse, Microchip Studio, MATLAB

EDUCATION

Marwadi University, Rajkot, Gujarat, India [NAAC A+ Accredited] **Sept 2022 - Present**
B.Tech. Information and Communication Technology (Specialization in VLSI and Embedded) *SGPA – 8.55*

GTS Vidhyala, Rajkot, Gujarat, India **May 2020 –April 2022**
Higher Secondary Schooling *Percentage – 50*

Shubham School, Rajkot, Gujarat, India **May 2019 –April 2020**
Matriculation *Percentage – 72*

VOLUNTEERING/ INDUSTRY VISIT/ ACHIEVEMENTS

- Volunteered in ATAL FDP 2025 organized by AICTE (All India Council for Technical Education) on **Lab-on-Chip**
- Industry visits at **eInfochips-An Arrow Company** at Ahmedabad, Gujarat
- Patent for Instrument Tuning Indian Patent Office **202521040536 A** Filed on 26/04/2025 and published on 16/05/2025

CERTIFICATES/COURSES

- Digital VLSI Testing **(NPTEL)**
- Lab Workshop on Embedded C and ARM Cortex **(NIELIT Calicut)**
- Hands-On Wireless Sensor Networks with ESP32 **(Infosys springboard)**